Name: Run Sokmeanheng
Class: Afternoon

SECP256k1 Literature Review

1. Definition of SECP256k1

SECP256k1 is an elliptic curve used in public-key cryptography and blockchain systems such as Bitcoin and Ethereum. It is defined by the Standards for Efficient Cryptography Group (SECG) and mainly supports the Elliptic Curve Digital Signature Algorithm (ECDSA).

2. Mathematical Foundation

SECP256k1 is defined over a finite prime field using the equation:
$y^2 = x^3 + 7 \pmod{p}$

Key Generation:
- Private key: random 256-bit number
- Public key: private key multiplied by a generator point

Digital Signatures:
Messages are hashed and signed with a private key and verified using the public key.

3. Why SECP256k1 Was Chosen for Bitcoin

- Faster computation
- Transparent parameters
- Strong security level
- Widely supported

4. Security and Efficiency

SECP256k1 provides about 128-bit security and is efficient compared to many other curves, especially in blockchain environments.

Curve	Speed	Trust Transparency	Usage
SECP256k1	Very Fast	High	Bitcoin, Ethereum
P-256 (NIST)	Slower	Lower	TLS, Government
Curve25519	Very Fast	High	Modern cryptography

5. Research Paper Review
Title: Guide to Elliptic Curve Cryptography by Hankerson, Menezes, and Vanstone explains ECC foundations and advantages.

6. Summarize the paper

The book Guide to Elliptic Curve Cryptography by Hankerson, Menezes, and Vanstone provides a comprehensive explanation of elliptic curve cryptography (ECC), including its mathematical principles, security assumptions, and practical implementations. The authors explain how ECC is based on the elliptic curve discrete logarithm problem, which is computationally infeasible to solve with current technology. The research demonstrates that ECC achieves the same level of security as traditional cryptographic systems such as RSA while using much smaller key sizes, resulting in faster computation and lower resource consumption. The book also emphasizes the importance of careful curve selection to prevent vulnerabilities. These findings are highly significant for blockchain security, as they justify the use of efficient and secure elliptic curves such as SECP256k1 for digital signatures and identity verification in decentralized systems.

References:

Hankerson, D., Menezes, A., & Vanstone, S. (2004).
Guide to Elliptic Curve Cryptography. Springer-Verlag, New York.